What Is Cybersecurity?

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Cybersecurity is the practice of protecting networks, systems, hardware and data from digital attacks. This guide walks you through the field of cybersecurity, including types of cyberattacks and its growing global importance in a digital world.

What Is Cybersecurity?

Cybersecurity is the practice of securing networks, systems and any other digital infrastructure from malicious attacks. With cybercrime damages costing an estimated $18.7 billion between 2017 and 2021, it's no wonder banks, tech companies, hospitals, government agencies and just about every other sector are investing in cybersecurity infrastructure to protect their business practices and the millions of customers that trust them with their data.

What's the best cybersecurity strategy? A strong security infrastructure includes multiple layers of protection dispersed throughout a company's computers, programs and networks. With so much at stake, it's not hyperbolic to think that cybersecurity tools and experts act as the last line of defense between our most vital information and digital chaos.

What Is Cybersecurity? | Video: edureka!

Cybersecurity Roles

There are cybersecurity professionals working across industries including healthcare, banking and government. They strive to identify vulnerabilities, fortify organizations against potential cyber risks and react to digital threats in real-time to protect critical systems and sensitive data.

Between May 2021 and April 2022, there were more than 1 million people employed in the cybersecurity workforce, according to CyberSeek, which maintains data about the cybersecurity job market. There were also over 700,000 cybersecurity-related job listings posted online during that same period. These were the sector's top job titles:

- Cybersecurity analyst
- Software developer
- Cybersecurity consultant
- Penetration and vulnerability tester
- Cybersecurity manager
- Network engineer
- Systems engineer
- Senior software developer
- Systems administrator

Types of Cyber Attacks

Cyber attacks come in all shapes and sizes. Some may be overt ransomware attacks (hijacking important business products or tools in exchange for money to release them), while some are covert operations by which criminals infiltrate a system to gain valuable data only to be discovered months after-the-fact, if at all. Criminals are getting craftier with their malicious deeds and here are some of the basic types of cyber attacks affecting thousands of people each day.

Types of Cybersecurity Threats

- Malware
- Phishing
- Social engineering
- Man-in-the-middle attack
- Zero-day attack

Malware

Malware is used to describe malicious software, including spyware, ransomware and viruses. It usually breaches networks through a vulnerability, like clicking on suspicious email links or installing a risky application. Once inside a network, malware can obtain sensitive information, further produce more harmful software throughout the system and can even block access to vital business network components (ransomware).

Phishing

Phishing is the practice of sending malicious communications, usually emails, designed to appear from reputable, well-known sources. These emails use the same names, logos and wording as a CEO or company to dull suspicions and get victims to click on harmful links. Once a phishing link is clicked, cyber criminals have access to sensitive data like credit cards, social security or login information.

Social Engineering

Social engineering is the process of psychologically manipulating people into divulging personal information. Phishing is a form of social engineering, where criminals take advantage of people's natural curiosity or trust. An example of more advanced social engineering is with voice manipulation. In this case, cyber criminals take an individual's voice (from sources like a voicemail or social media post) and manipulate it to call friends or relatives and ask for a credit card number or other personal information.

Man-in-the-Middle Attack

Man-in-the-Middle (MitM) attacks occur when criminals interrupt the traffic between a two-party transaction. For example, criminals can insert themselves between a public Wi-Fi and an individual's device. Without a protected Wi-Fi connection, cyber criminals can sometimes view all of a victim's information without ever being caught.

Zero-Day Attack

Zero-day attacks are becoming more and more common. Essentially, these attacks occur between a network vulnerability announcement and a patch solution. In the name of transparency and security, most companies will announce that they found a problem with their network safety, but some criminals will take this opportunity to unleash attacks before the company can come up with a security patch.

Further ReadingWhite Hat Hackers: Inside the World of Ethical Hacking

Cybersecurity Basics

A multi-layer cybersecurity approach is the best way to thwart any serious cyber attack. A combination of firewalls, software and a variety of tools will help combat malware that can affect everything from mobile phones to Wi-Fi. Here are some of the ways cybersecurity experts fight the onslaught of digital attacks.

Cybersecurity Best Practices

With cyber attacks occurring seemingly nonstop, firewalls, antivirus software, anti-spyware software and password management tools must all work in harmony to outwit surprisingly creative cybercriminals.

The U.S. Cybersecurity and Infrastructure Agency recommends individuals and organizations regularly update their software to ensure there are no vulnerabilities for potential attackers to take advantage of, as well as make sure they're using reputable up-to-date antivirus software. Other protective measures include changing default usernames and passwords, implementing multi-factor authentication, installing a firewall and identifying and avoiding suspicious emails.

Cybersecurity Best Practices

- Regularly updating software
- Using reputable, up-to-date antivirus software
- Changing default usernames and passwords
- Implementing multi-factor authentication
- Installing a firewall
- Identifying and avoiding suspicious emails

AI for Cybersecurity

Artificial intelligence is used in cybersecurity to thwart a wide variety of malicious cybercrimes. Security companies are training AI tools to predict data breaches, alert to phishing attempts in real-time and even expose social engineering scams before they become dangerous.

Securing Against Malware

Security against malware is certainly one of the most important issues today, and it will continue to be as malicious software evolves. An anti-virus software package is needed to combat any suspicious activity. These packages usually include tools that do everything from warning against suspicious websites to flagging potentially harmful emails.

Mobile Security

Mobile phones are one of the most at-risk devices for cyber attacks and the threat is only growing. Device loss is the top concern among cybersecurity experts. Leaving our phones at a restaurant or in the back of a rideshare can prove dangerous. Luckily, there are tools that lock all use of mobile phones — or enact multi-factor passwords — if this incident occurs.

Application security is also becoming another major issue. To combat mobile apps that request too many privileges, introduce Trojan viruses or leak personal information, experts turn to cybersecurity tools that will alert or altogether block suspicious activity.

Web Browser Security and the Cloud

Browser security is the application of protecting internet-connected, networked data from privacy breaches or malware. Anti-virus browser tools include pop-up blockers, which simply alert or block spammy, suspicious links and advertisements. More advanced tactics include two-factor authentication, using security-focused browser plug-ins and using encrypted browsers.

Wi-Fi Security

Using public Wi-Fi can leave you vulnerable to a variety of man-in-the-middle cyber attacks. To secure against these attacks, most cybersecurity experts suggest using the most up-to-date software and to avoid password-protected sites that contain personal information, such as banking, social media and email. Arguably, the most secure way to guard against a cyber attack on public Wi-Fi is to use a virtual private network, orVPN. VPNs create a secure network, where all data sent over a Wi-Fi connection is encrypted.